ELEARNING TUẦN 10: MACHINE LEARNING

1. How does Apache Spark's RDD (Resilient Distributed Dataset) abstraction facilitate the implementation of machine learning algorithms?
2. What are the advantages and disadvantages of using Apache Spark's MLlib library for machine learning compared to other popular libraries like TensorFlow or scikit-learn?
3. Can you explain the process of feature engineering and selection in the context of machine learning on Apache Spark?
4. How does Apache Spark handle data preprocessing tasks such as normalization, imputation, and feature scaling for machine learning applications?
5. What are some common machine learning algorithms available in Apache Spark's MLlib, and when would you choose one algorithm over another for a given task?
6. How does Apache Spark distribute computation across a cluster when training machine learning models, and what implications does this have for scalability and performance?
7. Can you discuss the role of hyperparameter tuning in optimizing machine learning models on Apache Spark, and what techniques or tools are available for this purpose?
8. How does Apache Spark support the training of deep learning models for tasks such as image recognition or natural language processing?
9. What are the challenges of training ensemble learning models on distributed systems like Apache Spark, and how can they be addressed?
10. Can you explain the concept of model serialization and deserialization in Apache Spark, and why it's important for deploying machine learning models?
11. How does Apache Spark handle class imbalance issues when training classification models, and what strategies can be employed to mitigate this problem?
12. What are the considerations for selecting appropriate evaluation metrics when assessing the performance of machine learning models trained on Apache Spark?
13. How does Apache Spark integrate with other data processing and storage systems such as Hadoop, Kafka, or Cassandra in the context of machine learning workflows?
14. Can you discuss some real-world applications where Apache Spark's machine learning capabilities have been particularly effective or transformative?
15. What are the limitations of Apache Spark's machine learning capabilities, and how might they impact the development and deployment of complex models?
16. How does Apache Spark handle streaming data for machine learning tasks, and what are the challenges associated with real-time model inference?
17. Can you compare and contrast the performance of Apache Spark's MLlib with other distributed machine learning frameworks such as Dask or Flink?
18. What are the best practices for monitoring and debugging machine learning pipelines on Apache Spark to ensure reliability and performance?
19. How does Apache Spark support model deployment and serving in production environments, and what considerations are important for maintaining scalability and reliability?

20. What advancements or future developments do you anticipate in Apache Spark's machine learning ecosystem, and how might they impact the landscape of distributed machine learning?